数据结构基础期中卷 cx
真题整理

一、判断题 (True or False)

R1-1. *(2 分)* Every node in the tree is the root of some subtree. (T/F)

R1-2. *(2 分)* If the five integers {1, 2, 3, 4, 5} are inserted into an initially empty binary search tree in any order starting with 3, the resulting trees will all have the same height.(T/F)

R1-3. *(2 分)* Given a max-heap with unique keys, if $i < j$, then the element p in level i must be larger than q in level j. (T/F)

R1-4. *(2 分)* Suppose that each single node in a disjoint set initially has height 0, the height of the resulting disjoint-set tree formed by performing union-by-height on n nodes will not exceed $\lfloor \log_2 n \rfloor$. (T/F)

R1-5. *(2 分)* If a complete binary tree has $n + 1$ nodes in the last level and n nodes in the second last level, then the total number of nodes in the tree must be $3n$. (T/F)

R1-6. *(2 分)* If N numbers are stored in a singly linked list in increasing order, then the average time complexity for binary search is $O(\log N)$. (T/F)

R1-7. *(2 分)* $114514^N = O(N!)$. (T/F)

二、单选题 (Multiple Choice)

R2-1. *(3 分)* Suppose we have a circular queue implemented on an array `a[20]`. We store the position of the first queue element in `front`, and the next position of the last queue element in `rear`. What is the size of the circular queue when `front==7` and `rear==3`?

A. 5

B. 4

C. 16

D. 17

R2-2. *(3 分)* For an in-order threaded binary tree, if the post-order and in-order traversal sequences are `FDEBGCA` and `FDBEACG`, respectively. Which pair of nodes' right links are both threads?

A. C and D.

B. C and F.

C. E and G.

D. B and G.

R2-3. *(3 分)* The array representation of the disjoint sets is given by {4, 5, 4, 5, -6, 9, -2, 7, -3, 9, 2}. Keep in mind that the elements are numbered from 1 to 11, and the depth of the root is 0. After invoking `Union(Find(6), Find(8)), Union(Find(10), Find(3))` with union-by-size and path compression, what is the maximum depth of all the nodes?

A. 4

B. 1

C. 3

D. 2

R2-4. *(3 分)* A "full tree" of degree 3 (every non-leaf node has 3 children) has 61 nodes. Then its height h is at most ____. Note: $h = 0$ for a single node tree.

A. cannot be determined

B. 21

C. 20

D. 19

R2-5. *(3 分)* Design an algorithm to determine whether the parentheses `()` in an expression are properly matched. Which data structure is the most suitable for this task?

A. Queue

B. Sequentially stored linear list

C. Heap

D. Stack

R2-6. *(3 分)* It is known that a 3-heap is a heap whose nodes have 3 children. In a min-3-heap with 2025 elements, which one of the following indices is NOT a possible position for the maximum key?

A. 676

B. 1012

C. 1013

D. 675

R2-7. *(3 分)* Consider the following operations performed on an initially empty Binary Search Tree (BST):

- Insert: 50, 30, 70, 20, 40, 60, 80
- Delete: 50 (replaced with its in-order successor)
- Insert: 55

After these operations, which node will be the in-order predecessor of 60 in the final BST?

A. The in-order predecessor does not exist for 60.

B. 40.

C. 55.

D. 30.

三、程序填空题 (Fill in the Blanks)

R5-1. *(9 分)* **Build MaxHeap** The function is to turn an array `A[]` of `N` elements into a max-heap.

```
#define leftchild(i) ( 2*(i)+1 )

void BuildMaxHeap( ElementType A[], int N )
{
    int i, j, child;
    ElementType Tmp;

    for ( i = (N-1)/2; i >= 0; i-- ) {
        j = i;
        for ( Tmp = A[j]; leftchild(j) < N; j = child ) {
            child = leftchild(j);
            if ( ________ (1) ________ )
                child ++;
            if ( ________ (2) ________ )    A[j] = A[child];
            else break;
```

```
        }
        ________ (3) ________;
    }
}
```

四、函数题 (Programming)

You are supposed to write a function of constructing a binary tree with given postorder and inorder sequences.

Format of Function

```
Tree BuildTree( int inorder[], int postorder[], int N );
```

where `Tree` is defined as the following:

```
typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree Left;
    Tree Right;
};
```

The inorder and postorder sequences are stored in `int inorder[]` and `int postorder[]`, respectively. The integer `N` is the number of nodes in the tree. The function `BuildTree` is supposed to return the pointer to the root of the constructed tree.

Sample Program of Judge

```
#include <stdio.h>
#include <stdlib.h>

#define MAXN 10

typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree Left;
    Tree Right;
};

Tree BuildTree( int inorder[], int postorder[], int N );

void Inorder_output( Tree T );   /* details omitted */
```

```
void Postorder_output( Tree T ); /* details omitted */

int main()
{
    Tree T;
    int inorder[MAXN], postorder[MAXN], N, i;
    scanf("%d", &N);
    for (i=0; i<N; i++) scanf("%d", &inorder[i]);
    for (i=0; i<N; i++) scanf("%d", &postorder[i]);
    T = BuildTree(inorder, postorder, N);
    printf("Check:\n");
    Inorder_output(T); printf("\n");
    Postorder_output(T); printf("\n");
    return 0;
}

/* Your function will be put here */
```

Sample Input

```
7
1 2 3 4 5 6 7
2 3 1 5 7 6 4
```

Sample Output

```
Check:
1 2 3 4 5 6 7
2 3 1 5 7 6 4
```

参考答案

一、判断题

R1-1. T

R1-2. T

R1-3. F

R1-4. T

R1-5. T

R1-6. F

R1-7. T

二、单选题

R2-1. C

R2-2. C

R2-3. C

R2-4. C

R2-5. D

R2-6. D

R2-7. C

三、程序填空题

R5-1 Build MaxHeap

(1) child + 1 < N && A[child] < A[child + 1]

(2) Tmp < A[child]

(3) A[j] = Tmp

四、函数题

```
Tree BuildTree( int inorder[], int postorder[], int N )
{
    if (N == 0) {
        return NULL;
    }

    Tree root = (Tree)malloc(sizeof(struct TreeNode));
    root->Element = postorder[N - 1];

    int k = 0;
    while (k < N && inorder[k] != root->Element) {
        k++;
    }

    root->Left = BuildTree(inorder, postorder, k);
    root->Right = BuildTree(inorder + k + 1, postorder + k, N - k - 1);
    return root;
}
```